# Chapter 8. Conditional Independence and Gaussian Graphical Models

Graphical models for discovering relations, Lecture 1 (partial correlation, precision matrix, GGM, graphical lasso, neighborhood selection)

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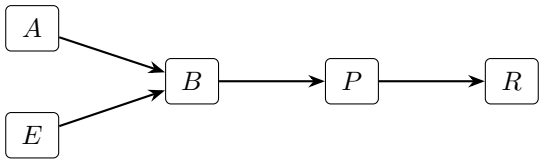
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## Roadmap of the causal-discovery part

| Lecture | Condition on the data | Relation to be discovered |
|---|---|---|
| 1 (Ch. 8) | Gaussian observational data | undirected graph of conditional relations |
| 2 (Ch. 9) | linear SEM with given directions | conditional independences a graph implies; limits of direction |
| 3 (Ch. 10) | all main variables observed | the equivalence class of DAGs that data can distinguish |
| 4 (Ch. 11) | hidden common causes allowed | latent confounding, relations and directions compatible with it |
| 5 (Ch. 12) | extra assumption on the error distribution | more specific directions inside the equivalence class |
| 6 (Ch. 13) | further identifiable models | directions for which causal discovery is possible |

### A running toy health example

The same five-variable health data are reused through all six lectures. The data-generating structure is




| | | | |
|---|---|---|---|
| $A$ | age | $E$ | amount of exercise |
| $B$ | body-mass index (BMI) | $P$ | blood pressure |
| $R$ | disease risk | | |

i.e. $E \to B \to P \to R$ and $A \to B$. Lecture 1 does not use directions and looks for conditional relations given the remaining variables; Lectures 2–3 study how directions are represented and how far they can be discovered; Lecture 4 adds a hidden lifestyle variable $U$, and Lecture 6 adds measurement error.

## 1 Correlation Mixes Direct and Indirect Relations

Let three standardized Gaussian variables be generated by

$$X_1 \sim N(0,1), \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.8X_2 + \epsilon_3, \qquad \epsilon_2, \epsilon_3 \overset{\text{ind}}{\sim} N(0, 0.36),$$

with $X_1, \epsilon_2, \epsilon_3$ mutually independent, so all three variances equal $0.64 + 0.36 = 1$.

| Pair | Correlation | Why the relation arises |
|---|---|---|
| $X_1, X_2$ | 0.8 | $X_1$ and $X_2$ are directly connected |
| $X_2, X_3$ | 0.8 | $X_2$ and $X_3$ are directly connected |
| $X_1, X_3$ | 0.64 | the relation is transmitted through $X_2$ |

Indeed $\mathrm{Cov}(X_1, X_3) = \mathrm{Cov}(X_1, 0.64X_1 + 0.8\epsilon_2 + \epsilon_3) = 0.64$. The right question is not "is there a correlation?" but "**does a relation remain after accounting for the other variables?**"

# 2 Conditional Independence

**Definition: conditional independence**

$X \perp\!\!\!\perp Y \mid Z$ if $p(x, y \mid z) = p(x \mid z)\, p(y \mid z)$ for all $z$, equivalently $p(x \mid y, z) = p(x \mid z)$: among cases with the same $Z$, knowing $Y$ gives no further information about $X$.

Marginal relation: $X_1 \not\perp\!\!\!\perp X_3$ — knowing $X_1$ changes what we know about $X_3$ (here Corr $= 0.64$). Conditional relation: $X_1 \perp\!\!\!\perp X_3 \mid X_2$ — comparing only cases with the same $X_2$, $X_1$ gives no additional information about $X_3$:

$$p(x_1, x_3 \mid x_2) = p(x_1 \mid x_2)\, p(x_3 \mid x_2).$$

Conditional independence does not fix causal direction, but it **narrows the candidate edges** of conditional dependence among variables.

# 3 Partial Correlation

## 3.1 Definition by residuals

Adjusting $X_1$ and $X_3$ for $X_2$:

$$X_1 = \alpha_1 + \beta_1 X_2 + r_1, \qquad X_3 = \alpha_3 + \beta_3 X_2 + r_3.$$

| Quantity | Meaning |
|---|---|
| $r_1$ | the part of $X_1$ not explained by $X_2$ |
| $r_3$ | the part of $X_3$ not explained by $X_2$ |
| $\mathrm{Corr}(r_1, r_3)$ | the partial correlation of $X_1, X_3$ given $X_2$ |

$$\rho_{13\cdot 2} = \mathrm{Corr}(r_1, r_3),$$

with population (linear-projection) residuals. For a set $S$, $\rho_{jk\cdot S}$ is defined in the same way after regressing both variables on $X_S$.

**Theorem: first-order partial correlation**

$$\rho_{13\cdot 2} = \frac{\rho_{13} - \rho_{12}\rho_{23}}{\sqrt{(1-\rho_{12}^2)(1-\rho_{23}^2)}}.$$

**Proof.** Standardize all variables. The projection coefficients are $\beta_1 = \rho_{12}$, $\beta_3 = \rho_{23}$, so $r_1 = X_1 - \rho_{12}X_2$ and $r_3 = X_3 - \rho_{23}X_2$. Then $\mathrm{Cov}(r_1, r_3) = \rho_{13} - \rho_{12}\rho_{23} - \rho_{23}\rho_{12} + \rho_{12}\rho_{23} = \rho_{13} - \rho_{12}\rho_{23}$, $\mathrm{Var}(r_1) = 1 - \rho_{12}^2$ and $\mathrm{Var}(r_3) = 1 - \rho_{23}^2$. ■

**Example: a correlation of 0.64 can disappear**

With $\rho_{12} = 0.8$, $\rho_{23} = 0.8$, $\rho_{13} = 0.64$,

$$\rho_{13\cdot 2} = \frac{0.64 - (0.8)(0.8)}{\sqrt{(1-0.8^2)(1-0.8^2)}} = 0.$$

Under joint Gaussianity we therefore conclude $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

# 4 The Precision Matrix

## 4.1 Zeros of the precision matrix are conditional independences

Let $X = (X_1, \dots, X_p)^\top \sim N_p(\mu, \Sigma)$ and call $\Omega = \Sigma^{-1}$ the **precision matrix**.

| Precision matrix | Interpretation |
|---|---|
| $\Omega_{jk} = 0$ | $X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$ |
| $\Omega_{jk} \neq 0$ | a relation between $X_j, X_k$ remains given all the other variables |

**Theorem: precision matrix and full-order partial correlation**

For $X \sim N_p(\mu, \Sigma)$ with $\Sigma \succ 0$ and $\Omega = \Sigma^{-1}$,

$$\rho_{jk\cdot -\{j,k\}} = -\frac{\Omega_{jk}}{\sqrt{\Omega_{jj}\Omega_{kk}}}, \qquad \text{so} \qquad \Omega_{jk} = 0 \iff X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}.$$

**Proof.** Let $A = \{j, k\}$ and $B$ the rest. For a Gaussian vector, $X_A \mid X_B$ is Gaussian with covariance equal to the Schur complement $\Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}$. By the block-inversion formula this Schur complement equals $(\Omega_{AA})^{-1}$, where $\Omega_{AA}$ is the $2 \times 2$ block of $\Omega$. Hence

$$\mathrm{Cov}(X_A \mid X_B) = \begin{pmatrix} \Omega_{jj} & \Omega_{jk} \\ \Omega_{jk} & \Omega_{kk} \end{pmatrix}^{-1} = \frac{1}{\Omega_{jj}\Omega_{kk} - \Omega_{jk}^2} \begin{pmatrix} \Omega_{kk} & -\Omega_{jk} \\ -\Omega_{jk} & \Omega_{jj} \end{pmatrix},$$

whose correlation is $-\Omega_{jk}/\sqrt{\Omega_{jj}\Omega_{kk}}$. The conditional covariance does not depend on $x_B$, so this is the partial correlation. For a bivariate Gaussian, zero correlation is equivalent to independence. ■

One matrix inversion gives, for every pair simultaneously, the relation after accounting for all other variables.

## 4.2 Turning the precision matrix into a graph: the GGM

**Definition: Gaussian graphical model**

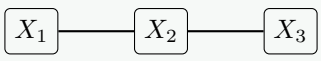
The GGM of $N_p(\mu, \Sigma)$ is the undirected graph on $\{1, \dots, p\}$ with

$$(j, k) \text{ is an edge} \iff \Omega_{jk} \neq 0.$$

**Example: the three-variable chain**

$$R = \begin{pmatrix} 1 & 0.8 & 0.64 \\ 0.8 & 1 & 0.8 \\ 0.64 & 0.8 & 1 \end{pmatrix}, \qquad \Omega = R^{-1} \approx \begin{pmatrix} 2.78 & -2.22 & 0 \\ -2.22 & 4.56 & -2.22 \\ 0 & -2.22 & 2.78 \end{pmatrix}.$$

Exactly: $\Omega_{11} = \Omega_{33} = 1/0.36$, $\Omega_{22} = 1.64/0.36$, $\Omega_{12} = \Omega_{23} = -0.8/0.36$, $\Omega_{13} = 0$ (check $R\Omega = I$). So the GGM is




The conditional relations $X_1 - X_2$ and $X_2 - X_3$ remain; $\Omega_{13} = 0$ removes the edge $X_1 - X_3$. The partial correlation of $X_1, X_2$ given $X_3$ is $0.8/\sqrt{1.64} \approx 0.62$, smaller than the marginal 0.8 (the same value follows from the first-order formula).

**Remark: from pairwise to global statements**

The definition uses the *pairwise* Markov property. For a positive density (e.g. a nondegenerate Gaussian) it is equivalent to the *global* Markov property: if a set $S$ separates node sets $A$ and $B$ in the graph (every path from $A$ to $B$ passes through $S$), then $X_A \perp\!\!\!\perp X_B \mid X_S$ (Hammersley–Clifford). In the chain, $\{X_2\}$ separates $X_1$ from $X_3$.

# 5 Estimating the Graph

## 5.1 Inverting the sample covariance produces spurious edges

With the sample covariance $S$, the simplest estimate is $\widehat{\Omega} = S^{-1}$.

| Data condition | Problem |
|---|---|
| finite sample | because of sampling error, entries that are truly 0 are estimated as small nonzero values |
| many variables | $S^{-1}$ is unstable and most entries are nonzero |
| $p \geq n$ | $S$ is singular and the inverse may not exist |

To discover relations we need a **sparse** precision-matrix estimate.

| Method | Core idea | Suitable when |
|---|---|---|
| partial-correlation tests | compute partial correlations from $S^{-1}$ and test $H_0 : \rho_{jk\cdot-\{j,k\}} = 0$ | $p < n$ and sample large enough |
| graphical lasso | $\ell_1$ penalty on the whole precision matrix, some entries estimated as 0 | a sparse positive-definite precision matrix is needed |
| neighborhood selection | Lasso regression of each variable on the rest; choose neighbors variable by variable | many variables, node-wise relations |
| CLIME | directly estimate a sparse $\Omega$ satisfying $S\Omega \approx I$ | estimating the precision matrix in high dimension |

The edge definition of the GGM is the same; the estimation method depends on the data conditions.

## 5.2 Testing a zero partial correlation

For Gaussian data, $H_0 : X_j \perp\!\!\!\perp X_k \mid X_S \iff H_0 : \rho_{jk\cdot S} = 0$. Let $\hat{\rho} = \hat{\rho}_{jk\cdot S}$ be the sample partial correlation and $q = |S|$.

| Test | Statistic | Reference distribution |
|---|---|---|
| partial-correlation $t$-test | $T = \hat{\rho}\sqrt{\dfrac{n-q-2}{1-\hat{\rho}^2}}$ | $t_{n-q-2}$ |
| Fisher-$z$ test | $Z = \sqrt{n-q-3}\,\operatorname{arctanh}(\hat{\rho})$ | approximately $N(0,1)$ |
| conditional distance correlation | $\mathrm{dCor}(X_j, X_k \mid X_{-\{j,k\}})$ | $p$-value by resampling |

**Derivation: where the degrees of freedom come from**

Regressing on $q$ variables plus an intercept uses $q+1$ degrees of freedom, so the residual correlation behaves like an ordinary sample correlation from $n-q$ observations. For an ordinary correlation from $m$ pairs, $T = r\sqrt{(m-2)/(1-r^2)} \sim t_{m-2}$ under $\rho = 0$,

and $\operatorname{arctanh}(r) \approx N(\operatorname{arctanh}\rho, 1/(m-3))$ by the delta method with variance stabilization ($\operatorname{Var}(r) \approx (1-\rho^2)^2/m$ and $\frac{d}{d\rho}\operatorname{arctanh}\rho = 1/(1-\rho^2)$). Substituting $m = n - q$ gives the table.

Failing to reject $H_0$ does *not* prove conditional independence.

## 5.3 Graphical lasso

**Derivation: Gaussian log-likelihood of $\Omega$**

For centered data, $\log p(x_i) = \frac{1}{2}\log\det\Omega - \frac{1}{2}x_i^\top\Omega x_i + c$, and $\sum_i x_i^\top\Omega x_i = \operatorname{tr}(\Omega\sum_i x_i x_i^\top) = n\operatorname{tr}(S\Omega)$. So the log-likelihood is $\frac{n}{2}\{\log\det\Omega - \operatorname{tr}(S\Omega)\} + c$, maximized at $\Omega = S^{-1}$ when $S \succ 0$.

The graphical lasso combines Gaussian fit with sparsity:

$$\widehat{\Omega}_\lambda = \arg\min_{\Omega\succ 0}\Big[\operatorname{tr}(S\Omega) - \log\det(\Omega) + \lambda\sum_{j\neq k}|\Omega_{jk}|\Big].$$

| Part of the objective | Role |
|---|---|
| $\operatorname{tr}(S\Omega) - \log\det(\Omega)$ | matches the covariance structure of the observed data |
| $\lambda\sum_{j\neq k}\lvert\Omega_{jk}\rvert$ | sets small off-diagonal entries exactly to 0 |

The objective is convex in $\Omega$; its optimality condition is $\Omega^{-1} - S - \lambda\Gamma = 0$ with $\Gamma_{jk} \in \partial|\Omega_{jk}|$ (and $\Gamma_{jj} = 0$). Consequently $|(\widehat{\Omega}^{-1})_{jk} - S_{jk}| \le \lambda$ for all $j \neq k$, with equality on the estimated edges. The solution is positive definite even when $p \ge n$. $\lambda$ is chosen by CV log-likelihood, EBIC or stability selection.

## 5.4 Neighborhood selection

**Theorem: conditional distribution of one node**

For centered $X \sim N_p(0, \Sigma)$ and $\Omega = \Sigma^{-1}$,

$$X_j \mid X_{-j} \sim N\Big(-\frac{\Omega_{j,-j}}{\Omega_{jj}}X_{-j},\ \frac{1}{\Omega_{jj}}\Big),$$

so the population coefficients of the regression of $X_j$ on the other variables are $\beta_k^{(j)} = -\Omega_{jk}/\Omega_{jj}$, and $\beta_k^{(j)} = 0 \iff \Omega_{jk} = 0$.

**Proof.** Apply the Schur-complement argument with $A = \{j\}$: $\operatorname{Var}(X_j \mid X_{-j}) = (\Omega_{jj})^{-1}$, and the conditional mean $\Sigma_{j,-j}\Sigma_{-j,-j}^{-1}X_{-j}$ equals $-\Omega_{jj}^{-1}\Omega_{j,-j}X_{-j}$ by the block-inverse identity $\Omega_{j,-j} = -\Omega_{jj}\Sigma_{j,-j}\Sigma_{-j,-j}^{-1}$. ■

In practice the population coefficients are estimated sparsely by the Lasso:

$$\hat{\beta}^{(j)} = \arg\min_\beta\Big\{\frac{1}{2n}\|X_j - X_{-j}\beta\|_2^2 + \lambda\|\beta\|_1\Big\}.$$

If $\hat{\beta}_k^{(j)} \neq 0$, $j - k$ is an edge candidate. In samples the two regressions ($j$ on $k$ and $k$ on $j$) may disagree, so they are combined by the **AND** rule (both nonzero) or the **OR** rule (at least one nonzero). Each regression costs one Lasso, which parallelises over nodes and scales to large $p$; the price is that the result is a graph, not a positive-definite $\widehat{\Omega}$.

### 5.5 CLIME

$$\widehat{\Omega} = \arg\min_{\Omega} \|\Omega\|_1 \quad \text{subject to} \quad \|S\Omega - I\|_\infty \leq \lambda,$$

a linear program that splits into one column at a time; the estimate is symmetrized afterwards. It estimates the sparsest matrix that is approximately an inverse of $S$.

## 6 GGM Edges Are Conditional Associations, Not Causal Directions

| Interpretable from a GGM | Not interpretable from a GGM alone |
|---|---|
| Does a relation remain after accounting for the other observed variables? | Which variable is the cause and which the effect? |
| What are the sign and size of the partial correlation? | Is an edge a direct causal effect? |
| Which variables are conditionally connected to many others? | Is there a hidden common cause? |

For example, $X_j - X_k$ means that the two variables remain related after accounting for the remaining observed variables. It means neither $X_j \to X_k$ nor $X_j \leftarrow X_k$. Worse, conditioning on a common *effect* can create an edge between two independent causes (Ch. 9). To infer directions we need DAGs and additional structure-learning assumptions.

## Summary

- Correlation mixes direct and indirect relations; conditional independence asks what remains given the other variables.
- Partial correlations are correlations of residuals; for Gaussians, $\Omega_{jk} = 0 \iff X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$ and $\rho_{jk\cdot\text{rest}} = -\Omega_{jk}/\sqrt{\Omega_{jj}\Omega_{kk}}$.
- The GGM puts an edge where $\Omega_{jk} \neq 0$; it is estimated by partial-correlation tests, graphical lasso, neighborhood selection or CLIME.
- GGM edges are undirected conditional associations; directions require the DAG machinery of the next chapters.